# Experimental setup

## A. Simulation harness — USED FOR ALL COMMITTED RESULTS

### Unmodified firmware logic

smart_home.ino compiled against host stubs for Arduino, FreeRTOS, WiFi, MQTT, NVS and the OLED. The firmware source is not altered for testing.

### Campaign

6,000 trials
seed base 1000
rates 0 to 20,000 msg/s
fork() per trial

### Fuzzer

3,000,000 iters
seed 7, 6 classes
ASan + UBSan
oracle: no state change, no counter

### Limitation that defines the open step

The harness advances a logical clock. There is no radio, no true parallel execution and no CPU contention, so latency variance is zero.

### Analysis

analyse.py: aggregation, Mann-Whitney U with tie correction, standard library only. make_figures.py: validated palette.

## B. Hardware rig — SPECIFIED, NOT YET RUN

### Device under test

ESP32 DevKit
running v2.2.0
unmodified

GPIO ▶

### Measuring board

second ESP32,
independent clock,
logic probe

### Four conditions

| | | |
|---|---|---|
| **A.** | idle (control) | no RF isolation |
| **B.** | legitimate network load | no RF isolation |
| **C.** | command-interface stress | no RF isolation |
| **D.** | radio-layer stress | AUTHORISED ISOLATED RF ONLY |

### Dependent variable

Physical sensor-to-siren latency distribution, compared across conditions with the same non-parametric test (analyse_hw.py).

**A to C need two boards and a probe. No attack tooling is publishe**